

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO1: Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1,BL2
Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:20

Annexure A SHORT ANSWER TEST

Criteria	Conceptual Understanding(3)	Accuracy of Answer(2.5)	Application of Knowledge(2)	Completeness of Response(1.5)	Clarity & Presentation(1)	Total(10)
Weightage	30%	25%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						

Code of Conduct:

I SK Sameer (192370288) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SK. Sameer

Short Answer Test

1. Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.

- (a) Enumerate three classes of schemas that are popularly used for modeling data warehouses.
 - (b) Draw a schema diagram for the above data warehouse using one of the schema.
 - (c) To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge).
- (5 Marks)

2. Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (5 Marks)

- (a) Draw a snowflake schema diagram for the data warehouse.
- (b) Starting with the base cuboid [student, course, semester, instructor], what specific OLAP operations (e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses for each Big University student.
- (c) If each dimension has five levels (including all), such as "student < major < status < university < all", how many cuboids will this cube contain (including the base and apex cuboids)?

3. Design a data warehouse for a regional weather bureau. The weather bureau has about 1,000 probes, which are scattered throughout various land and ocean locations in the region to collect basic weather data, including air pressure, temperature, and precipitation at each hour. All data are sent to the central station, which has collected such data for over 10 years. Your design should facilitate efficient querying and on-line analytical processing, and derive general weather patterns in multidimensional space. (5 Marks)

4. A data cube, C , has n dimensions, and each dimension has exactly p distinct values in the base cuboid. (5 Marks)

Assume that there are no concept hierarchies associated with the dimensions.

- (a) What is the maximum number of cells possible in the base cuboid?
- (b) What is the minimum number of cells possible in the base cuboid?
- (c) What is the maximum number of cells possible (including both base cells and aggregate cells) in the data cube, C ?
- (d) What is the minimum number of cells possible in the data cube, C ?

CO1 - AT1

- (a) Star schema, snowflake schema, and fact Constellation (Galaxy) schema are the three commonly used data warehouse schemas, they organize data for efficient storage and analysis.
- b) A star schema has one central fact table containing count and charge, connected to Time, Doctor, and patient dimension tables.
- c) An SQL query uses the SUM() function and Group By clause to calculate the total count and total charge for each doctor, patient, and time period.
9. (a) A snowflake schema stores normalized dimension tables such as student, course, semester, and instructor connected to a central fact table.
- b) Use Roll-up from semester to year, slice for the CS course, and Average aggregation to obtain the required grades.
- c) if each dimension has 5 levels, the total number of cuboids is $5^4 = 625$.

3. Design a Data warehouse for a Regional weather Bureau.

A weather data warehouse is designed to store and analyze weather information collected from nearly 1000 probes located across land and ocean regions. The main dimensions are Time (hour, day, month, year), Location (region, state, station), and weather station (probe). The measurements include air pressure, temperature, and precipitation. The warehouse stores historical weather data for over 10 years, enabling fast OLAP operations such as roll-up, drill-down, slice, and dice.

It helps meteorologists analyze climate trends, generate weather forecasts, compare seasonal patterns, and support better decision-making through efficient multidimensional analysis.

a) Maximum number of cells in cuboid.

The maximum number of cells in the base cuboid is p^n , where p is the number of values in each dimension and n is the number of dimension.

b) Minimum number of cells in the base cuboid.

The minimum number of cells in the base cuboid is also p^n because every combination of dimension values must be represented.

c) Maximum number of cells in the data cube.

The maximum number of cells in the complete data cube, including all aggregate is $(p+1)^n$.

d) Minimum number of cells in data cube.

The minimum number of cells in the complete data cube is 2^n , representing the smallest number of aggregated cuboids.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand